

Difference between su - and sudo - on a Linux system

Short answer: su - switches your entire shell to another user (usually root) and loads that user's full login environment. sudo - is not a real command—sudo uses flags like sudo -i or sudo -u— but the *closest equivalent* is sudo -i, which gives you a root login shell *without* requiring the root password.

Core difference at a glance

- **su -** — You *become* the target user (often root) and load their login environment. Requires **root's password**.
- **sudo -i** — You get a root login shell using **your own password**, if permitted by sudoers.

What su - actually does

su - (or su --login) starts a **full login shell** as the target user:

- Loads the target user's environment variables (PATH, HOME, SHELL, etc.)
- Changes to the target user's home directory
- Requires the **target user's password**
- Gives you a persistent root session until you exit

This is equivalent to logging in as root directly.

What the “sudo equivalent” really is: sudo -i

There is **no sudo - command**, but sudo -i is the functional counterpart to su -:

- Starts a **root login shell**
- Loads root's login environment
- Uses **your password**, not root's
- Logs the action for auditing
- Ends when you exit the shell

This is safer because you never need the root password, and every privileged action is logged.

Side-by-side comparison

Feature	<code>su -</code>	<code>sudo -i</code>
Password required	Root's password	Your password
Privilege scope	Full root shell	Full root shell
Environment	Root's login environment	Root's login environment
Logging	No command logging	Logged to auth logs
Security model	High risk (full root session)	Safer (sudoers rules + audit trail)
Typical use	Traditional Unix admin	Modern Linux (Ubuntu, Debian, etc.)

Security implications

- `su -` is riskier: once you're root, **every command is dangerous**, and nothing is logged.
- `sudo -i` is safer: it enforces sudoers policy and logs privileged actions.

When to use which

- Use `sudo -i` when you need a temporary root login shell but want modern security practices.
- Use `su -` only when:
 - You *must* use the root password
 - You're on a system without `sudo`
 - You're doing traditional Unix administration